

ICD 1.6/1.10[ab]

“A&G to NIRI/GNIRS On Instrument Wavefront Sensors”

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Hubert Yamada
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_____ Systems Engineering _____

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Chapter 1 Introduction

1–1.0 Description

1–1.1 Purpose

This Interface Control Document (ICD) serves the following purposes:

- To describe the commands and parameters that will pass between the A&G system and the On-Instrument Wavefront Sensor (OIWFS) Components Controller (CC).
- To provide a prototype which may, with minimal modification, be used to describe the GNIRS OIWFS, and the OIWFS for any system which copies the NIRI OIWFS hardware and software.

This document only describes the ways in which the NIRI/GNIRS OIWFS differs from the prototypical OIWFS which is described in ICD 1.6/1.10 [19] and in [20].

At some time in the future, the NIRI/GNIRS OIWFS information may be merged with ICD 1.6/1.10.

In addition to the documents mentioned in the previous paragraphs, the specific interfaces described in this document are governed by the following parent Interface Control Documents, which describe the general properties of particular kinds of interface:

- ICD 1a [24], which describes how the OCS deals with commands and attributes and the behavior the OCS expects from the systems it commands.
- ICD 1b [25], which gives a general description of the way the Observatory Control System communicates with the other principal systems. It is also related to ICD 7a, [27], which describes the way an Instrument Control System communicates with its subsystems.
- ICD 2 [26], which describes the way in which an Instrument Control System makes information available to the outside world (and the OCS) by means of “Status Information Records”.
- ICD 7b [28], which describes the way that the Telescope Control System (TCS) interacts with its subsystems. (From a control standpoint, the OIWFS is a subsystem of the A&G, which is a subsystem of the TCS.)

The intended audience for this document is:

- The NIRI reviewers;
- The GNIRS reviewers;
- Any other groups which will be using the design of the NIRI/GNIRS OIWFS;
- The developers of the Gemini Acquisition and Guidance units;
- The developers of the GNIRS OIWFS components controller.

1–1.2 Scope

This document describes only aspects of the NIRI/GNIRS OIWFS software system which differ from, or are more specific than, the Gemini standard on-instrument wavefront sensor

(OIWFS), described in [19]. It is assumed that the reader is familiar with both of those documents.

This document does *not* describe the following topics:

- The OIWFS hardware and electronics (see [11] and [12], summarized in [3]);
- The OIWFS detector software, hardware, and electronics (see [12] and [13]);
- The GNIRS software, hardware, and electronics;
- Other instruments which will be using the NIRI/GNIRS OIWFS;
- The OIWFS Interlock system (see [6]);
- The names of the NIRI OIWFS mechanism positions (see [37]);
- The names of the GNIRS OIWFS mechanism positions;
- Interfacing an instrument to the NIRI/GNIRS OIWFS (see [6]);
- The software design (see [2], and [3]).

Unless otherwise stated, any references to the OIWFS refers only to the OIWFS mechanisms. The OIWFS detector is a completely separate system, and does not have an EPICS interface.

1–2.0 Related Documents and Drawings

1–2.1 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

Chapter 2 Details of the A&G to NIRI/GNIRS OIWFS Interface

2-1.0 Introduction

2-2.0 Physical System Interfaces

2-2.1 Mechanical Interface

Not Applicable for a software ICD.

2-2.2 Optical Interface

Not Applicable for a software ICD.

2-2.3 Electronic Interface

2-2.3.1 NIRI

The system hardware architecture for the NIRI OIWFS control software is the same as described in [16], with the following exception:

- The Instrument Sequencer, Components Controller, and OIWFS Components Controller will share the same IOC.

There will also be the following electronic VME modules:

- The XVME-240 Digital I/O Board:
 - To monitor the status of the stepper-motor control modules;
 - To reset the stepper-motor control modules;
- Four custom-built VME sensor-support boards:
 - To read out the NIRI OIWFS Hall-effect sensors;
- Two XVME-566 Analog Input Modules:
 - To read out the VME sensor-support boards for the NIRI OIWFS Hall-effect sensors;
- Two Oregon Microsystems VME8-8 Stepper-motor controller boards
 - To control the NIRI OIWFS stepper-motors.

Note that these boards will be shared between the NIRI OIWFS hardware and the NIRI hardware, and will perform other functions that are not listed above. In particular, the XVME-240 board is shared between the NIRI CC, the NIRI OIWFS, and is also used to interface to the interlock bus. See [3] for more details.

2-2.3.2 GNIRS / Standalone OIWFS

The system hardware architecture for the GNIRS OIWFS control software is the same as described in [16], with the following exception:

- The OIWFS Components Controller and its supporting hardware will be in the same VME crate as other GNIRS hardware, but will have a separate bus.

There will also be the following electronic VME modules:

- The XVME-240 Digital I/O Board;
 - To monitor the status of the stepper-motor control modules;
 - To reset the stepper-motor control modules;
- One custom-built VME sensor-support board:
 - To read out the GNIRS OIWFS Hall-effect sensors;
- One XVME-566 Analog Input Modules:
 - To read out the VME sensor-support boards for the GNIRS OIWFS Hall-effect sensors;
- One Oregon Microsystems VME8-8 Stepper-motor controller boards
 - To control the GNIRS OIWFS stepper-motors.

These boards will not be shared between the OIWFS hardware and any other system; however, the XVME-240 board will be used to interface to the interlock bus.

2-2.4 Mass/Balance

Not Applicable for a software ICD.

2-2.5 Thermal Interface

Not Applicable for a software ICD.

2-3.0 Software/Control Function Interface

2-3.1 Overview

2-3.1.1 Communications infrastructure

The communications infrastructure is the same as described in [16].

2-3.1.2 Software architecture

The software architecture is described in [2].

2-3.2 Behavior

The behavior of the NIRI/GNIRS OIWFS/A&G interface is the same as described in [20] and [19].

2-3.3 Implementation

This section contains tables summarizing the EPICS records used for the interface. The next sections (Section 2-3.4 on page 17 and Section 2-3.5 on page 21) contain the details of what those records do.

It is assumed in all cases that the record names given in this document will be prefixed by the string “niri:” or “nirs:”¹, as appropriate when the database is implemented, except for the SAD database which will be prefixed by “niri:”, “niri::”, “nirs:”, or “nirs::”².

2–3.3.1 Command (CAD/CAR) records

The columns of the command record tables are as follows:

Description. A brief description of the command.

Attribute. The name of the attribute which the Observatory Control System associates with a particular parameter of a particular command. Component position names are listed separately in Table 7.

CAD Record / Cad Record.FIELD. The name of the EPICS (minus the “niri:” or “nirs:” prefix) together with the field in that CAD record associated with each parameter of the command.

CAD ordering number. The recommended order in which this command should be executed. CAD records should be connected to their associated APPLY record in the order 1, 2, 3, 4, 5, etc... CAD records within the same ordering number set can be connected to the APPLY record in any order in that set.

CAR record. The CAR record associated with the command. Note that several CAD records may share the same CAR record.

2–3.3.1.1 CAD/CAR records—System Commands

These commands affect the overall behavior of NIRI and the NIRI OIWFS; they affect the GNIRS OIWFS, but not the other GNIRS components. Note, in particular, that it is not possible to reboot NIRI and the NIRI OIWFS separately. It may be possible to reboot the GNIRS OIWFS without rebooting GNIRS.

TABLE 1. NIRI/GNIRS system engineering command CAD/CAR records

Description	Attribute	CAD Record / CAD Record.FIELD (“niri:” or “nirs:” +)	CAR record
Reboot.		wfs:reboot	N/A
Select debug mode.	Debug mode	wfs:debug.A	wfs:debugC
Flush data from internal buffers.		wfs:flush	wfs:flushC

1. Or some other Gemini-assigned name, which must be at most four alpha-numeric characters, followed by a colon.
2. Or some other Gemini-assigned name, which must be at most four alpha-numeric characters, followed by one or two colons.

2-3.3.1.2 OIWFS command CAD/CAR records—OIWFS components

The commands in Table 2 are the only ones that are used during routine observation.

TABLE 2. OIWFS command CAD/CAR records

Description	Attribute ^a	CAD Record / CAD Record.FIELD ("niri:" or "nirs:" +)	CAR record ("niri:" or "nirs:" +)
Select a filter in the filter wheel ^b .	filter name	wfs:filtSel.A	wfs:filtC
Park the filter wheel.	(none)	wfs:filtPark	
Move gimbal mirrors, to given (x, y) position and leave it stationary. (x, y) is the position in the focal plane measured in mm in the instrument's frame of reference.	x y	wfs:prbMove.A wfs:prbMove.B	wfs:prbC
Park the gimbal mirror.	(none)	wfs:prbPark	
Select the focus-stage position by name.	position name	wfs:focSel.A	wfs:focC
Set the focus-stage position in mm of translation.	position	wfs:focMove.A	
Park the focus stage.	(none)	wfs:focPark	
The gimbal and focus stage are caused to follow a continuous stream of updated (x=gimbal x, y=gimbal y, z=focus) positions. These positions are not defined as CAD record arguments but in separate records (see ICD 1.6/1.10). ^c	(none)	wfs:folFollow	folC
"Follow" mode, initiated by the wfs:folFollow command, is terminated.	(none)	wfs:folStop	
Set the tolerance values for all three axes (x=gimbal x, y=gimbal y, z=focus) in mm. The tolerance is used to calculate the wfs:folInToler SIR record.	x y z	wfs:folSetTol.A wfs:folSetTol.B wfs:folSetTol.C	folSetC

a. "(none)" means that there are no arguments.

b. ICD 1.6/1.10 [19] has three separate components: a field-stop, a neutral-density filter, and a color filter. For the NIRI/GNIRS OIWFS, these are combined into a single wheel which may have multiple optical elements at each position.

c. The focus is not designed to be operated continuously, and it should not be necessary to do so. By default, the stream of focus offset values is ignored.

2-3.3.1.3 OIWFS Engineering commands CAD/CAR records—OIWFS components

The following commands are used for engineering purposes, and are not expected to be used during normal observing. Note that, although these commands are connected to the OIWFS APPLY record, they are normally activated directly by the engineering user interface.

The following commands affect the overall behavior of the OIWFS but do not affect the science detector.

TABLE 3.

NIRI/GNIRS OIWFS engineering command CAD/CAR records

Description	Attribute	CAD Record / CAD Record.FIELD ("niri:" or "nirs:" +)	CAR record ("niri:" or "nirs:" +)
Datum filter wheel.		wfs:filtDatm	wfs:filtC
Set filter wheel position in position numbers ^a .	position	wfs:filtMove.A	
Set filter wheel position in engineering units (motor microsteps) ^b .	motor position	wfs:filtEngMove.A	
Confirm that filter wheel has not lost its datum position.		wfs:filtRdtm	
Confirm that filter wheel and its Hall-effect sensors are operational		wfs:filtDiag	
Datum the gimbal mirror.		wfs:prbDatm	wfs:prbC
Datum the x-axis of the gimbal mirror.		wfs:prbDatm	wfs:prbxC
Tip gimbal mirror along its x-axis, given x position in engineering units (motor microsteps).	motor position	wfs:prbxEngMove.A	
Confirm that the OIWFS gimbal mirror x-axis motor has not lost its datum position.		wfs:prbxRdtm	
Confirm that the OIWFS gimbal mirror x-axis and its Hall-effect sensors are operational		wfs:prbxDiag	
Datum the y-axis of the gimbal mirror.		wfs:prbDatm	wfs:prbyC
Tip gimbal mirror along its y-axis, given y position in engineering units (motor microsteps).	motor position	wfs:prbyEngMove.A	
Confirm that OIWFS gimbal mirror y-axis motor has not lost its datum position.		wfs:prbyRdtm	
Confirm that OIWFS gimbal mirror y-axis motor and its Hall-effect sensors are operational		wfs:prbyDiag	
Datum focus mechanism.		wfs:focDatm	wfs:focC
Set focus stage translation in engineering units (motor microsteps).	motor position	wfs:focEngMove.A	
Confirm that focus stage has not lost its datum position.		wfs:focRdtm	
Confirm that focus stage and its Hall-effect sensors are operational		wfs:focDiag	

a. Only the standard positions (excluding datum and park) are fully repeatable.

b. Only the standard positions (excluding datum and park) are fully repeatable.

2-3.3.2 Status Information Records

The following tables show the status information provided by the OIWFS. The tables have the following columns:

SIR record. The part of the name of the EPICS SIR record which follows the "niri:" or "nirs:" prefix³.

Units. The units in which the quantity is stored. (Ustep refers to units of motor microsteps; “user units” or “real-world units” refer to the arbitrary units described by the Units SIR variable.)

Comments. A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.

2-3.3.2.1 WFS Status Information

Table 4 contains status information for all of the WFS.

TABLE 4. The WFS Overall Status Information

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS included	Type	Units	Comments
wfs:debugMode	WSDMODE	Never	string		Debugging mode [NOLOG NONE MIN FULL]
wfs:health		Never	string		WFS Health [GOOD WARNING BAD]
wfs:heartBeat	WSHRTB	Never	long	count	Heart beat record: Shows that WFS software is alive by returning a constantly increasing value.
wfs:historyLog	WSHLOG	Never	array of char ^a		History log record ^b
wfs:issPort	WSISSPT	Start	long	number	ISS port on which instrument is installed
wfs:label	WSLABEL	Never	string		The FDSC field describes the WFS Components Controller
wfs:name	WSNAME	Never	string		WFS Components Controller Name
wfs:simMode	WSSMODE	Start	string		Simulation mode [NONE FULL FAST VSM]
wfs:state	WSSTATE	Never	string		Inst. state [BOOTING INITIALIZING RUNNING CONFIGURING]
wfs:wfsBeam	WSWFSBM	Start	long	0/1	Is the WFS detector unobstructed? [YES=0 NO=1] ^c

- Should be “bigstring”
- This record is not useful, because of various EPICS limitations.
- The wfs:wfsBeam record only indicates whether the OIWFS components controller mechanisms are obstructing the beam; it does not indicate that light is actually getting to the detector. Furthermore, it does not account for the obstruction caused by the pick-off mirror. The instrument needs to combine the wfs:wfsBeam with information about any other components in order to provide an instrument-wide wfsBeam record.

- Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI/GNIRS database(s) (for example “niri:” or “nirs:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:” or “nirs:”.

2-3.3.2.2 The Wavefront Sensor Components Status Information

The following table summarizes the information for the various components.

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:filtDatumed	WSFLDTM	Never	long	0/1	Is the filter wheel datumed? [YES=0 NO=1]
wfs:filtEngCyclic	WSFLECY	Never	long	0/1	Ignore limits for the filter wheel? [YES=0 NO=1]
wfs:filtEngMax	WSFLEMX	Never	long	motor microsteps	Maximum position of the filter wheel in engineering units (motor microsteps)
wfs:filtEngMin	WSFLEMN	Never	long	motor microsteps	Minimum position of the filter wheel in engineering units (motor microsteps)
wfs:filtEngName	WSFLENM	Start	string		Engineering name of the filter wheel position
wfs:filtEngPos	WSFLEPS	Start	long	motor microsteps	Position of the filter wheel in engineering units (motor microsteps)
wfs:filtLabel	WSFLLBL	Never	string		The FDSC field describes the filter wheel
wfs:filtMax	WSFLMAX	Never	double	position number	Maximum position of the filter wheel in real-world units (position number)
wfs:filtMin	WSFLMIN	Never	double	position number	Minimum position of the filter wheel in real-world units (position number)
wfs:filtName	WSFLNAM	Start	string		User name of the filter wheel position
wfs:filtParked	WSFLPRK	Never	long	0/1	Is the filter wheel parked? [YES=0 NO=1]
wfs:filtPos	WSFLPOS	Start	double	position number	Position of the filter wheel position in real-world units (position number)
wfs:filtRdtmVal	WSFLRDT	Never	long	motor microsteps	Returned value from the filter wheel redatum command
wfs:filtReject	WSFLREJ	Never	long	0/-1	Combined the filter wheel CAR state [GOOD=0 REJECT=-1]
wfs:filtState	WSFLSTE	Never	string		the filter wheel SNL state
wfs:filtUnits	WSFLUNT	Start	string		Definition of real-world units ("user units") for the filter wheel ("pos num").
wfs:focAgDiff	WSFOCADF	Never	double	mm	The focus stage offset error in real-world units (mm of translation)
wfs:focAgEngDiff	WSFOCAEF	Never	double	motor microsteps	The focus stage offset error in engineering units (motor microsteps)
wfs:focDatumed	WSFOCDTM	Never	long	0/1	Is the focus stage datumed? [YES=0 NO=1]
wfs:focEngCyclic	WSFOCECY	Never	long	0/1	Ignore limits for the focus stage? [YES=0 NO=1]
wfs:focEngMax	WSFOCEMX	Never	long	motor microsteps	Maximum position of the focus stage in engineering units (motor microsteps)
wfs:focEngMin	WSFOCEMN	Never	long	motor microsteps	Minimum position of the focus stage in engineering units (motor microsteps)

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:focEngName	WSFOCENM	Start	string		Engineering name of the focus stage position ^a
wfs:focEngPos	WSFOCEPS	Start	long	motor microsteps	Position of the focus stage in engineering units (motor microsteps)
wfs:focInToler	WSFOCINT	Never	long	0/1	Is the focus stage offset in tolerance ^b ? [YES=0 NO=1]
wfs:focLabel	WSFOCLBL	Never	string		The FDSC field describes the focus stage
wfs:focMax	WSFOCMAX	Never	double	mm	Maximum position of the focus stage in real-world units (mm of translation)
wfs:focMin	WSFOCMIN	Never	double	mm	Minimum position of the focus stage in real-world units (mm of translation)
wfs:focName	WSFOCNAM	Start	string		User name of the focus stage position ^c
wfs:focParked	WSFOCPRK	Never	long	0/1	Is the focus stage parked? [YES=0 NO=1]
wfs:focPos	WSFOCPOS	Always	double	mm	Position of the focus stage position in real-world units (mm of translation)
wfs:focRdtmVal	WSFOCRDT	Never	long	motor microsteps	Returned value from the focus stage redatum command
wfs:focReject	WSFOCREJ	Never	long	0/-1	Combined focus stage CAR state [GOOD=0 REJECT=-1]
wfs:focState	WSFOCSTE	Never	string		the focus stage SNL state
wfs:focTol	WSFOCTOL	Never	double	mm	Tolerance of the focus stage offset in real-world units (mm of translation)
wfs:focUnits	WSFOCUNT	Always	string		Definition of real-world units ("user units") for the focus stage. ("mm").
wfs:folDatumed	WSFLDTM	Never	long	0/1	Are all dynamic mechanisms datumed? [YES=0 NO=1]
wfs:folInToler	WSFOLINT	Never	long	0/1	Are all dynamic mechanisms in tolerance ^d ? [YES=0 NO=1]
wfs:folIsFollow	WSFLFOL	Never	long	0/1	Are all dynamic components in follow mode? [YES=0 NO=1]
wfs:folIsIdle	WSFLIDL	Never	long	0/1	Are all dynamic components idle? [YES=0 NO=1]
wfs:folLabel	WSFLLBL	Never	string		The FDSC field describes the dynamic mechanisms
wfs:folName	WSFLNAM	Start	string		Dynamic components: name of current position
wfs:folParked	WSFLPRK	Never	long	0/1	Are all dynamic mechanisms parked? [YES=0 NO=1]
wfs:folPresent	WSFLPRS	Never	double	seconds	Current Gemini time
wfs:folReject	WSFLREJ	Never	long	0/1	Combined reject state for Follow Command
wfs:prbDatumed	WSPBDTM	Never	long	0/1	Is/Are Gimbal Mirror datumed? [YES=0 NO=1]

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:prbLabel	WSPBLBL	Never	string		The FDSC field describes the gimbal mirror
wfs:prbName	WSPBNAM	Start	string		the gimbal mirror name of current position ^e
wfs:prbParked	WSPBPRK	Never	long	0/1	Is the gimbal mirror parked? [YES=0 NO=1]
wfs:prbReject	WSPBREJ	Never	long	0/1	Combined reject state for the gimbal mirror
wfs:prbSky:xPos	WSPBSXPS	Never	double	sky units	Position of the gimbal mirror x in sky units
wfs:prbSky:yPos	WSPBSYPS	Never	double	sky units	Position of the gimbal mirror y in sky units
wfs:prbSkyLabel	WSPBSLBL	Never	string		The FDSC field describes the gimbal mirror
wfs:prbSkyUnits	WSPBSUNT	Start	string		Definition of "sky units"
wfs:prbxAgDiff	WSPBXADF	Never	double	mm	The gimbal-mirror x-axis offset error in real-world units (mm in the focal plane)
wfs:prbxAgEngDiff	WSPBXAEF	Never	double	motor microsteps	The gimbal-mirror x-axis offset error in engineering units (motor microsteps)
wfs:prbxDatumed	WSPBXDTM	Never	long	0/1	Is the gimbal-mirror x-axis datumed? [YES=0 NO=1]
wfs:prbxEngCyclic	WSPBXECY	Never	long	0/1	Ignore limits for the gimbal-mirror x-axis? [YES=0 NO=1]
wfs:prbxEngMax	WSPBXEMX	Never	long	motor microsteps	Maximum position of the gimbal-mirror x-axis in engineering units (motor microsteps)
wfs:prbxEngMin	WSPBXEMN	Never	long	motor microsteps	Minimum position of the gimbal-mirror x-axis in engineering units (motor microsteps)
wfs:prbxEngName	WSPBXENM	Start	string		Engineering name of the gimbal-mirror x-axis position ^f
wfs:prbxEngPos	WSPBXEPS	Start	long	motor microsteps	Position of the gimbal-mirror x-axis in engineering units (motor microsteps)
wfs:prbxInToler	WSPBXINT	Never	long	0/1	Is the gimbal-mirror x-axis offset in tolerance ^g ? [YES=0 NO=1]
wfs:prbxLabel	WSPBXLBL	Never	string		The FDSC field describes the gimbal-mirror x-axis
wfs:prbxMax	WSPBXMAX	Never	double	mm	Maximum position of the gimbal-mirror x-axis in real-world units (mm in the focal plane)
wfs:prbxMin	WSPBXMIN	Never	double	mm	Minimum position of the gimbal-mirror x-axis in real-world units (mm in the focal plane)
wfs:prbxName	WSPBXNAM	Start	string		User name of the gimbal-mirror x-axis position ^h
wfs:prbxParked	WSPBXPRK	Never	long	0/1	Is the gimbal-mirror x-axis parked? [YES=0 NO=1]

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:prbxPos	WSPBXPOS	Always	double	mm	Position of the gimbal-mirror x-axis position in real-world units (mm in the focal plane)
wfs:prbxRdtmVal	WSPBXRDT	Never	long	motor microsteps	Returned value from the gimbal-mirror x-axis redatum command
wfs:prbxReject	WSPBXREJ	Never	long	0/-1	Combined the gimbal-mirror x-axis CAR state [GOOD=0 REJECT=-1]
wfs:prbxState	WSPBXSTE	Never	string		the gimbal-mirror x-axis SNL state
wfs:prbxTol	WSPBXTOL	Never	double	mm	Tolerance of the gimbal-mirror x-axis offset in real-world units (mm in the focal plane)
wfs:prbxUnits	WSPBXUNT	Always	string		Definition of real-world units ("user units") for the gimbal-mirror x-axis. ("mm").
wfs:prbyAgDiff	WSPBYADF	Never	double	mm	The gimbal-mirror y-axis offset error in real-world units (mm in the focal plane)
wfs:prbyAgEngDiff	WSPBYAEF	Never	double	motor microsteps	The gimbal-mirror y-axis offset error in engineering units (motor microsteps)
wfs:prbyDatumed	WSPBYDTM	Never	long	0/1	Is the gimbal-mirror y-axis datumed? [YES=0 NO=1]
wfs:prbyEngCyclic	WSPBYECY	Never	long	0/1	Ignore limits for the gimbal-mirror y-axis? [YES=0 NO=1]
wfs:prbyEngMax	WSPBYEMX	Never	long	motor microsteps	Maximum position of the gimbal-mirror y-axis in engineering units (motor microsteps)
wfs:prbyEngMin	WSPBYEMN	Never	long	motor microsteps	Minimum position of the gimbal-mirror y-axis in engineering units (motor microsteps)
wfs:prbyEngName	WSPBYENM	Start	string		Engineering name of the gimbal-mirror y-axis position ⁱ
wfs:prbyEngPos	WSPBYEPS	Start	long	motor microsteps	Position of the gimbal-mirror y-axis in engineering units (motor microsteps)
wfs:prbyInToler	WSPBYINT	Never	long	0/1	Is the gimbal-mirror y-axis offset in tolerance? [YES=0 NO=1]
wfs:prbyLabel	WSPBYLBL	Never	string		The FDSC field describes the gimbal-mirror y-axis
wfs:prbyMax	WSPBYMAX	Never	double	mm	Maximum position of the gimbal-mirror y-axis in real-world units (mm in the focal plane)
wfs:prbyMin	WSPBYMIN	Never	double	mm	Minimum position of the gimbal-mirror y-axis in real-world units (mm in the focal plane)
wfs:prbyName	WSPBYNAM	Start	string		User name of the gimbal-mirror y-axis position ^k

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:prbyParked	WSPBYPRK	Never	long	0/1	Is the gimbal-mirror y-axis parked? [YES=0 NO=1]
wfs:prbyPos	WSPBYPOS	Always	double	mm	Position of the gimbal-mirror y-axis position in real-world units (mm in the focal plane)
wfs:prbyRdtmVal	WSPBYRDT	Never	long	motor microsteps	Returned value from the gimbal-mirror y-axis redatum command
wfs:prbyReject	WSPBYREJ	Never	long	0/-1	Combined the gimbal-mirror y-axis CAR state [GOOD=0 REJECT=-1]
wfs:prbyState	WSPBYSTE	Never	string		the gimbal-mirror y-axis SNL state
wfs:prbyTol	WSPBYTOL	Never	double	mm	Tolerance of the gimbal-mirror y-axis offset in real-world units (mm in the focal plane)
wfs:prbyUnits	WSPBYUNT	Always	string		Definition of real-world units ("user units") for the gimbal-mirror y-axis ("mm").

- a. For the focus stage the position returned by wfs:focEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- b. Valid only when the system is in follow mode.
- c. For the focus stage the position returned by wfs:focName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- d. Valid only when the system is in follow mode.
- e. For the gimbal mirror, the position returned by wfs:prbName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- f. For the gimbal mirror x-axis, the position returned by wfs:prbxEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- g. Valid only when the system is in follow mode.
- h. For the gimbal mirror x-axis, the position returned by wfs:prbxName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- i. For the gimbal mirror y-axis, the position returned by wfs:prbyEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- j. Valid only when the system is in follow mode.
- k. For the gimbal mirror y-axis, the position returned by wfs:prbyName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.

2-3.3.2.3 Component Health Information

The following table summarizes the health information records. All records return one of the strings: “GOOD”, “WARNING”, or “BAD”. Each record describes one aspect of the health of a component

TABLE 6. OIWFS Components Controller Component Health Information

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Included	Type	Units	Comments
wfs:fltDatumHealth		Never	string		The health of the filter wheel datum state
wfs:fltHallHealth		Never	string		The health of the filter wheel Hall-effect sensors
wfs:fltHealth		Never	string		The overall health of the filter wheel
wfs:fltModuleHealth		Never	string		The health of the filter wheel driver
wfs:fltPosErrHealth		Never	string		The health of the filter wheel backlash correction
wfs:fltSnIHealth		Never	string		The health of the filter wheel state code
wfs:focDatumHealth		Never	string		The health of the focus stage datum state
wfs:focHallHealth		Never	string		The health of the focus stage Hall-effect sensors
wfs:focHealth		Never	string		The overall health of the focus stage
wfs:focModuleHealth		Never	string		The health of the focus stage driver
wfs:focPosErrHealth		Never	string		The health of the focus stage backlash correction
wfs:focSnIHealth		Never	string		The health of the focus stage state code
wfs:folHealth		Never	string		The overall health of the dynamic mechanisms (the gimbal mirror and focus stage)
wfs:prbHealth		Never	string		The overall health of the gimbal mirror
wfs:prbxDatumHealth		Never	string		The health of the gimbal-mirror x-axis datum state
wfs:prbxHallHealth		Never	string		The health of the gimbal-mirror x-axis Hall-effect sensors
wfs:prbxHealth		Never	string		The overall health of the gimbal mirror x-axis
wfs:prbxModuleHealth		Never	string		The health of the gimbal-mirror x-axis driver
wfs:prbxPosErrHealth		Never	string		The health of the gimbal-mirror x-axis backlash correction
wfs:prbxSnIHealth		Never	string		The health of the gimbal-mirror x-axis state code
wfs:prbyDatumHealth		Never	string		The health of the gimbal-mirror y-axis datum state
wfs:prbyHallHealth		Never	string		The health of the gimbal-mirror y-axis Hall-effect sensors
wfs:prbyHealth		Never	string		The overall health of the gimbal mirror y-axis
wfs:prbyModuleHealth		Never	string		The health of the gimbal-mirror y-axis driver
wfs:prbyPosErrHealth		Never	string		The health of the gimbal-mirror y-axis backlash correction
wfs:prbySnIHealth		Never	string		The health of the gimbal-mirror y-axis state code

2-3.3.3 Position names

Table 7 contains a list of valid position names. Note that these names are case sensitive and must be entered exactly as presented here, including underscores. In addition to the listed

positions, all mechanisms respond to the special names “datum” and “park” which may or may not be distinct from the listed positions.

TABLE 7. Position Names (observing commands)

Component	Position Names	Engineering Position ^a	Comments
Filter Wheel	See [37]		
Gimbal Mirror	Neutral	(comp)	
	Minimum	(comp)	
	Maximum	(comp)	
Gimbal Mirror X-Axis	Neutral	pos03	Mirror centered
	Minimum	pos02	Minimum travel location
	Maximum	pos00	Maximum travel location
Gimbal Mirror Y-Axis	Neutral	pos03	Mirror centered
	Minimum	pos02	Minimum travel location
	Maximum	pos00	Maximum travel location
Focus Stage	Neutral	pos00	Center of travel
	Minimum	pos01	Minimum travel location
	Maximum	pos02	Maximum travel location

a. “(comp)” means that the positions are mapped into component positions, and not sent directly to a mechanism.

2–3.4 Detailed Command Description

A general table of the commands accepted by the NIRI/GNIRS OIWFS, together with their arguments, is given in Table 1 on page 7. This section contains a detailed description of each of the commands that will be sent between the A&G and the NIRI/GNIRS control software. It is based on the required behavior of the commands specified in [21] and [32].

If a directive is accepted it will cause the NIRI/GNIRS OIWFS to change its configuration. The “applyC” CAR record will go “BUSY” while the OIWFS control software is changing its configuration and go to “IDLE” if completed successfully or to “ERR” if the configuration change fails.

2–3.4.1 Sequence Commands

2–3.4.1.1 INIT

On receipt of this command, the OIWFS control software will execute its initialization sequence, in which it will: read its hardware setup files and lookup tables, and reset itself to the start-up state. The “initC” CAR record will go “BUSY” while the initialization is being carried out and go to “IDLE” if completed successfully or to “ERR” if the initialization fails. No mechanisms will be moved.

2–3.4.1.2 DATUM

The OIWFS will locate the datum (i.e. index, home or reference point) for all its mechanisms. The “datumC” CAR record will go “BUSY” while the datum search is being carried out and go to “IDLE” if completed successfully or to “ERR” if the datum searched fails.

2-3.4.1.3 PARK

On receipt of this command, the OIWFS will move its mechanisms to a configuration in which it can safely be switched off. It will utilize all the low-level CAD records dedicated to parking the individual mechanisms. The “parkC” CAR record will go “BUSY” while the mechanisms are being parked and go to “IDLE” if completed successfully or to “ERR” if any mechanism fails to park successfully.

2-3.4.1.4 DIAGNOSE

On receipt of this command, the OIWFS will perform self tests to ensure that its subsystems are healthy. The “diagnoseC” CAR record will go “BUSY” while the tests are being carried out and go to “IDLE” if completed successfully or to “ERR” if the tests fail.

Note: unlike the TEST command, the DIAGNOSE command will move mechanisms, as necessary to diagnose problems.

2-3.4.2 General Engineering Commands

2-3.4.2.1 REBOOT

On receipt of this command, the OIWFS control software will reboot its IOC and go through the same start-up procedure it would normally do when first switched on. The working system will reboot into the simulation mode NONE and debug mode NONE. The simulator will reboot into simulation mode FULL and debug mode NONE. There is no CAR record associated with this command because connection with the IOC will be lost while it reboots.

This command will also reboot the NIRI CC and IS. It will not reboot the GNIRS CC and IS.

2-3.4.2.2 DEBUG

On receipt of this command, the OIWFS control software will change to the given debug mode (NONE, MIN or FULL). This mode determines the amount and frequency of information logged by the NIRI control software, as described in the “Debugging Modes” section of ICD 1a, [24]. The “debugC” CAR record will go briefly “BUSY” and then back to “IDLE”.⁴

2-3.4.3 OIWFS Standard Mechanisms

The OIWFS will have a single filter wheel, a detector focusing stage, and a two-axis gimbal mirror. The filter wheel is considered a “discrete” mechanisms, because it normally is moved only to one of a small number of locations. The focus stage and both axes of the gimbal mirror are considered continuous mechanisms, because they can move to a wide range of locations.

The following commands will act on all of the above mentioned mechanisms; however, as noted below, the use of certain commands with certain mechanisms is discouraged.

2-3.4.3.1 Sel—Select Position by Name

This is the principal interface for all continuous mechanisms. It is provided as a convenience for discrete mechanisms. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

4. Note: This was the required behavior for this system. This requirement was dropped, so the debug command is non-functional.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Translate desired position name into an engineering position with the aid of a look-up table.
 Look up the expected value for the Hall-effect sensors at the desired position location.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism to the engineering position corresponding to the desired position.
 Verify that the expected sensor value matches the measured value.

2-3.4.3.2 Move—Move to specified Position

This is the principal interface for all discrete components. It is available for continuous components, but is of limited use for normal use of the instrument⁵. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the focus mechanism to the engineering position corresponding to the desired focus.
 For some mechanisms, compute the expected values for the Hall-effect sensors.
 For some mechanisms, verify that the final hall effect sensors match the expected values.

2-3.4.3.3 Park—Park Mechanism

This command is used to prepare a mechanism for instrument shutdown. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism to its parking position.
 Verify that the expected sensor value matches the measured value.

The park command is not needed for most OIWFS components, and is provided for uniformity. The single exception is the focus stage. The focus stage should be parked while the instrument being cooled down or warmed up.

2-3.4.3.4 Datm—Datum mechanism

This command is used to find the datum (home) position which defines the mechanism locations. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

5. For discrete mechanisms, the units are position numbers; i.e., the first position is at 1.0, the second position is at 2.0.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism and search for its datum.
 Set the hardware datum position.

2-3.4.3.5 MoveEng—Set position in engineering units (motor microsteps)

This command is used to move the components to locations using units that are convenient for working directly with the hardware. Note that for multi-axis components (e.g., the steering mirrors), the command acts only on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism to the specified engineering position.

The Hall-effect sensors will not be used for confirmation.

2-3.4.3.6 Rdtm—Redatum mechanism.

This command provides a quick test that will detect a loss of positioning. It will detect small positioning errors, but can underestimate large positioning errors. Note that for multi-axis components (e.g., the steering mirrors), the command acts only on the individual mechanisms. It returns an estimate of the difference between the actual datum position and the current datum position (which can occur, e.g., if a mechanism is sticking, being driven too quickly, or being driven with too high an acceleration).

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism and search for its datum.

The hardware datum position will be left unchanged. This command returns a value in the RdtmVal record for this mechanism.

This command is not supported by the pupil viewer or the window cover, because those mechanisms use a home-finding algorithm which is incompatible with this command.

2-3.4.3.7 Diag—Diagnose mechanism

This is a quick test that will detect a loss of positioning. It will detect arbitrarily large positioning errors, but will fail to detect small positioning errors. Note that for multi-axis components (e.g., the steering mirrors), command acts only on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure that an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism and attempt to verify that the motors and Hall-effect sensors are operating.

If an error is detected, the diagnose command will fail and will return an error code in its CAR record.

2-3.5 Detailed Status information

2-3.5.1 General records

2-3.5.1.1 name

This record contains a string constant describing the name of the instrument. For NIRI this record will contain “NIRI WFS”; for GNIRS the record will contain “NIRS WFS”.

2-3.5.1.2 state

This record will take the values BOOTING, INITIALISING, RUNNING and CONFIGURING and reflect the state of the OIWFS control software while it is initializing. The activities corresponding to these states are

- BOOTING — the state during IOC Init and while any database default values are being set.
- INITIALISING — the OIWFS control software will be checking its own hardware and reading its hardware configuration files during this phase.
- RUNNING — the OIWFS control software is ready to accept commands.
- CONFIGURING — the OIWFS control software is ready to accept commands but it is busy configuring and unable to make observations.

2-3.5.1.3 health

This record will reflect the overall health of the OIWFS. Expected values are GOOD, WARNING or BAD. If the overall health is BAD then the OIWFS will be unusable. If the Health is set to WARNING then the OIWFS will be able to execute some functions but not perhaps to specification.

2-3.5.1.4 historyLog

The record used to contain messages to be recorded in the history log.

2-3.5.2 Component Status Records

2-3.5.2.1 Datumed

This record indicates that a component has been datumed. For multi-axis components, this flag will only be set if all mechanisms have been datumed. This record takes on the following values:

- 0 = Datumed
- 1 = Not Datumed

2-3.5.2.2 EngCyclic

This record indicates that a mechanism may be set to a location beyond its physical limits (convenient for circular mechanisms like filter wheels). Note that this only affects the higher level commands; the low-level software may choose to ignore any invalid requests. This record takes on the following values:

- 0 = Limits can be exceeded
- 1 = Limits may not be exceeded

This record is for internal use by the OIWFS software. Its value should not be changed. It is documented here only for completeness.

2-3.5.2.3 EngMax

This record contains the maximum position that the mechanism can be moved to, in engineering units (motor microsteps).

2-3.5.2.4 EngMin

This record contains the minimum position that the mechanism can be moved to, in engineering units (motor microsteps).

2-3.5.2.5 EngName

This record contains the engineering name of the current mechanism position. This represents a physical configuration of a component, and will never change (unlike the “Name” record which may change, if filters or other optical elements are replaced or rearranged). The names of the positions are:

- park;
- home;
- pos00, pos01, pos02, ... , up to the maximum number of positions;
- INVALID (if not in a standard location).

2-3.5.2.6 EngPos

This record contains the current position of a mechanism, in engineering units (motor microsteps).

2-3.5.2.7 Label

The FDSC field of this record contains a text string which describes a component. This is convenient in providing a title string for edd/dm screens.

2-3.5.2.8 Max

This record contains the maximum position that the mechanism can be moved to, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.9 Min

This record contains the maximum position that the mechanism can be moved to, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.10 Name

This record contains the name of the current component position. This name typically describes a filter or other optical element and will be redefined if filters or other optical elements are replaced or rearranged (unlike the “EngName” record which describes a component’s physical configuration). The current names are listed in [36].

Note that for multi-axis components, there will be a Name record for each individual mechanism, as well as for the component as a whole.

2-3.5.2.11 Parked

This record indicates that a component is in a parked state. For multi-axis components, this record will be set only if all mechanisms are parked. This field takes the values:

- 0 = All mechanisms parked;
- 1 = Not all mechanisms are parked.

2-3.5.2.12 Pos

This record contains the current position of a mechanism, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.13 RdtmVal

This record contains the result of the last redatum command, in engineering units (motor microsteps). (The redatum command returns an estimate of the how far the current mechanism datum position is from the actual datum position; it is useful in diagnosing hardware problems.)

2-3.5.2.14 Reject

This record contains a combination of the values of all CAD records for all mechanisms that make up a component. It contains the following values:

- -1 = At least one CAD record is in a reject state;
- 0 = No CAD records are in a reject state.

This record exists as a convenience for the user interface. There is no reason for the OCS to use it. It is documented here only for completeness.

2-3.5.2.15 State

This record contains a string which describes the current state of the mechanism-support code.

This record is for debugging the OIWFS software. The possible values that this record may take may change, without notice. It is documented here only for completeness.

2-3.5.2.16 Units

This record contains a text string which describes the real world units (e.g., “mm” or “arc sec”)

2-3.5.3 Component Health Records

All of these records return the standard health values: “GOOD”, “WARNING”, and “BAD”.

2-3.5.3.1 DatumHealth

This record reports “WARNING” if a mechanism (or if any mechanism of a multi-axis component) is not datumed. It is cleared by successful execution of a datum command.

2-3.5.3.2 HallHealth

This record reports “WARNING” if a Hall-effect sensor has been disabled. (The sensors can only be disabled from the engineering user interface, and are normally disabled only to deal with a failed sensor.)

This record reports “BAD” if both sensors of a Hall-effect sensor pair have been disabled. Note that a mechanism is still usable if both sensors are disabled, but cannot be datumed. (The sensors can only be disabled from the engineering user interface, and are normally disabled only to deal with a failed sensor.)

2-3.5.3.3 Health

This record reports the most severe error value of all other health records.

2-3.5.3.4 ModuleHealth

This record reports “BAD” if the stepper-motor driver module has entered a fault state. This normally indicates a serious hardware problem. It may be possible to clear the error state by issuing a driver-reset from the engineering user interface. If it is not possible to clear the error, then the stepper-motor driver module or other related hardware has failed, and must be repaired.

2-3.5.3.5 PosErrHealth

This record reports “WARNING” if a mechanism (or if any mechanism of a multi-axis component) has been moved in such a way that proper backlash correction has not been made. In this case, the reported mechanism position may not be the actual mechanism position. This situation can arise if a movement command has been canceled while a motion is in progress; if a movement command failed to complete correctly, if certain low-level engineering commands are executed, and immediately after the control software has been rebooted. It is cleared by successful execution of any motion command.

2-3.5.3.6 SnlHealth

This record reports “BAD” or “WARNING” if the underlying mechanism support code detects an error while a mechanism is in motion. The only situation which is likely to arise is “WARNING”, if an operation takes longer than expected.

2-3.6 Alarm conditions

The following situations will generate an alarm.

- Hall effect sensor disabled
- Motor-control module fault
- Mechanism not datumed
- Mechanism not backlash corrected
- Problem detected by SNL code (time-out)
- Temperature changing
- Temperature out of range

2-3.7 Debugging and Maintenance

This interface can be tested and debugged by means of an engineering user interface supplied as part of the OIWFS.

2-3.7.1 Debugging Modes

The OIWFS control software will have the standard debugging modes described in ICD 1a, [24].

2-3.7.2 Calibration

If a hall sensor fails or if a mechanism must be disassembled for maintenance, the sensors will need to be recalibrated. A separate program is provided for this purpose. Calibration procedures are described in a separate document [5].

2-3.8 Simulation

The Gemini simulation modes are described in ICD 1a, [24]. The OIWFS control software supports only the “NONE” mode. The simulator supports only the “FULL” and “FAST” modes.

The “FAST” simulation mode removes most of the simulated delays which are built into the “FULL” simulation mode, but does not fully meet the Gemini specification for the FAST simulation mode. The “VSM” mode is not available.

To switch from operating mode to simulation mode requires an alternate start-up script to be selected, and a reboot (see the release notes [7]). The command to switch between the “FAST” and “FULL” simulations modes is an engineering-only command and does not fall within the scope of this document (see [8]).

2-4.0 Safety Issues

It will be necessary to avoid moving OIWFS components while the mechanical components are not at thermal equilibrium. Records are provided to provide a temperature interlock. Details are described in [6].

